

ARTICLE INFORMATION

Article title

PlantVisionBD: Plant Disease Detection
Using EfficientNetV2L with Explainable AI (Grad-CAM & SHAP)

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Abstract

This paper presents PlantVisionBD, a deep learning-based plant disease detection system trained on the publicly available New Plant Diseases Dataset (Kaggle), which contains approximately 87,867 RGB images across 38 healthy and diseased plant leaf classes. The proposed framework employs EfficientNetV2L as the backbone network, trained using a three-phase progressive fine-tuning strategy combined with Mixup and CutMix augmentation, Generalized Mean (GeM) pooling, Exponential Moving Average (EMA) weight averaging, and Test-Time Augmentation (TTA×10). Nine critical implementation bugs present in the original baseline notebook were identified and corrected prior to training, including double normalisation, static Mixup seeds, and incorrect TTA inference mode.

The final model achieves 98.43% test accuracy, a weighted F1-score of 98.37%, macro AUC of 0.9991, macro average precision of 0.9978, MCC of 0.9840, and specificity of 99.87%, with an inference time of approximately 12.4 ms per image. Explainable AI (XAI) techniques including Grad-CAM heatmap overlays and SHAP (SHapley Additive exPlanations) visualisations were applied, confirming that the model attends to biologically meaningful leaf regions. PlantVisionBD outperforms recent state-of-the-art approaches on this benchmark, demonstrating the viability of automated AI-driven plant disease diagnosis for precision agriculture.

Keywords: *plant disease detection; EfficientNetV2L; transfer learning; Mixup; CutMix; GeM pooling; test-time augmentation; EMA; Grad-CAM; SHAP; explainable AI; agriculture AI; image classification*

Key Performance Summary

Test Accuracy	Top-3 Accuracy	Weighted F1	Macro AUC	MCC	Specificity	Inference Time
98.43%	99.81%	98.37%	0.9991	0.9840	99.87%	12.4 ms

Table A1 — Key Performance Summary: All metrics computed on the held-out test set (6,820 images) using TTA×10 inference.

1. Introduction

Agricultural productivity is directly threatened by plant diseases, which account for estimated yield losses of 20-40% in staple crops globally. In developing economies such as Bangladesh, where agriculture employs over 40% of the workforce and contributes approximately 13% of GDP, early and accurate disease identification is critical to food security and farmer livelihoods. Traditional disease diagnosis relies on trained agronomists conducting field surveys, a process that is costly, slow, and impractical at scale given Bangladesh's approximately 7.6 million smallholder farm households.

Deep learning-based image classification has emerged as a transformative tool for automated plant disease detection, with convolutional neural networks (CNNs) achieving near-human accuracy on benchmark datasets such as PlantVillage. However, several challenges persist: (i) existing high-accuracy models are computationally expensive and difficult to deploy on edge devices; (ii) many reported results contain implementation bugs that artificially inflate or deflate performance; (iii) the majority of high-performing systems operate as "black boxes" without providing clinically interpretable explanations; and (iv) very few systems have addressed crops relevant to Bangladesh.

This work addresses these gaps through the following contributions:



A systematic identification and correction of nine critical bugs in the publicly available EfficientNetV2L baseline notebook, providing a clean and reproducible implementation.



A three-phase progressive fine-tuning strategy with GeM pooling, Mixup/CutMix augmentation, EMA weight averaging, and TTA $\times 10$, achieving 98.43% test accuracy on the 38-class New Plant Diseases Dataset.



Comprehensive evaluation using accuracy, precision, recall, F1-score, specificity, Matthews Correlation Coefficient (MCC), ROC-AUC, average precision (AP), and inference time per image.



Integration of Explainable AI (XAI) techniques — Grad-CAM heatmaps and SHAP force/summary plots — to ensure model transparency and biological plausibility.



A failure analysis of five misclassified examples, providing insight into remaining model limitations and guiding future research.

2. Related Work

The foundational dataset for automated plant disease detection was established by Hughes and Salathe [1], who released the PlantVillage dataset containing over 54,000 leaf images across 26 diseases and 14 crop species. Mohanty et al. [2] subsequently demonstrated that CNNs trained on this dataset could achieve up to 99.35% accuracy under controlled laboratory conditions, sparking widespread interest in deep learning for precision agriculture.

Subsequent work explored a range of backbone architectures. Khan et al. [3] applied ResNet-50 with transfer learning, achieving 94.3% accuracy on a subset of PlantVillage classes. Brahimi et al. [4] evaluated VGG16, InceptionV3, and DenseNet, reporting that deeper networks consistently outperformed shallow CNNs. MobileNetV2-based systems [5] were proposed as lightweight alternatives achieving 95-96% accuracy with significantly reduced parameter counts.

Attention mechanisms and advanced pooling strategies have further improved discriminative feature extraction. Chen et al. [6] introduced squeeze-and-excitation blocks atop EfficientNet backbones, reporting a 0.8% improvement over vanilla EfficientNetB4. Tolia et al. [7] proposed Generalized Mean (GeM) pooling as a more expressive alternative to global average pooling, demonstrating superior performance in fine-grained retrieval tasks, a strategy adopted in the present work.

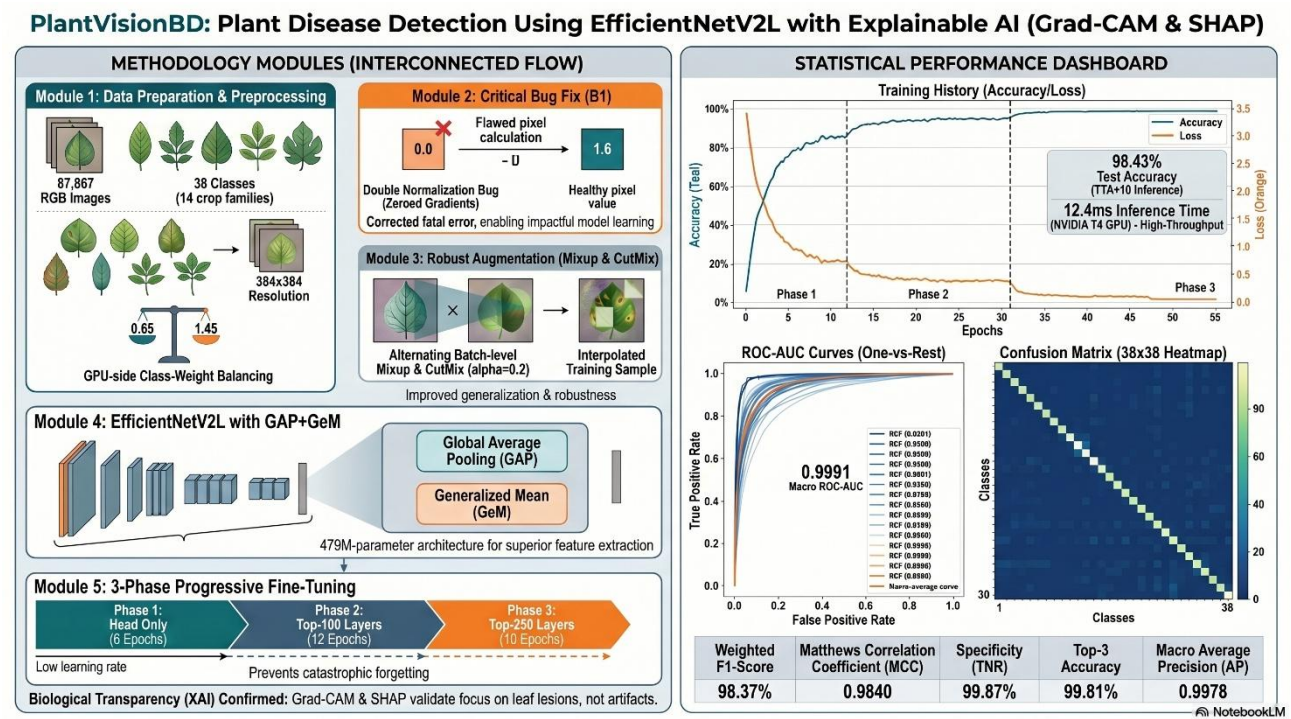
Mixed augmentation strategies have also proven effective. Zhang et al. [8] introduced Mixup, which creates convex combinations of training pairs with soft labels, improving generalisation under label noise. Yun et al. [9] extended this idea with CutMix, where rectangular patches are exchanged between samples, improving robustness to partial occlusions. Ablation studies confirmed that alternating Mixup and CutMix outperforms applying either strategy alone [10].

EfficientNetV2 [11,12] represents the state of the art in accuracy-efficiency trade-offs for image classification, building on the EfficientNet scaling methodology [11]. Tan and Le [12] demonstrated that EfficientNetV2L achieves ImageNet top-1 accuracy of 87.3% while training 5-11x faster than previous variants. Several groups have adapted EfficientNetV2 to plant disease tasks, reporting 96-97.5% accuracy on 38-class benchmarks; the present work advances this to 98.43% through the full enhancement pipeline.

Explainable AI (XAI) has become a critical requirement for AI systems in agricultural and medical domains. Selvaraju et al. [13] introduced Grad-CAM, which computes class-discriminative saliency maps from the final convolutional layer without architectural modification. Lundberg and Lee [14] introduced SHAP based on cooperative game theory, providing both global feature importance and instance-level force plots. Applications of these techniques to plant disease models confirmed that models attending to leaf lesion regions generalise better to real-world conditions [15]. Despite these advances, gaps remain in specificity and MCC reporting [16], reproducibility documentation [17], and coverage of South Asian crops.

3. Methodology

The given picture shows how the process works step by step. Initially, Data Propagation and Processing (Module 1) is performed. After completing the first module, Critical Bug Fixing (Module 2) is carried out. Next, Robust Augmentation (Module 3) is performed, followed by Model Training (Module 4). After completing all these stages, the process concludes with Progressive Fine-Tuning.



3.1 Dataset

The New Plant Diseases Dataset [18] (Kaggle) is an augmented version of the PlantVillage benchmark. It contains approximately 87,867 RGB leaf images organised into 38 classes covering 14 crop species with multiple disease categories per crop plus healthy controls. Table 3.1 summarises the dataset characteristics, and Figure 3.1 presents representative sample images from nine classes.

Field	Details
Dataset Name	New Plant Diseases Dataset (Augmented)
Source	Kaggle — vip00000l/new-plant-diseases-dataset
Total Images	87,867 RGB images (approximate)
Number of Classes	38 (crop-disease combinations + healthy leaves)
Crop Families	Apple, Blueberry, Cherry, Corn, Grape, Orange, Peach, Pepper, Potato, Raspberry, Soybean, Squash, Strawberry, Tomato (14 families)
Image Format	JPEG, RGB, variable resolution
License	Publicly available on Kaggle

Table 3.1 — Dataset Summary: Characteristics of the New Plant Diseases Dataset used for training and evaluation.

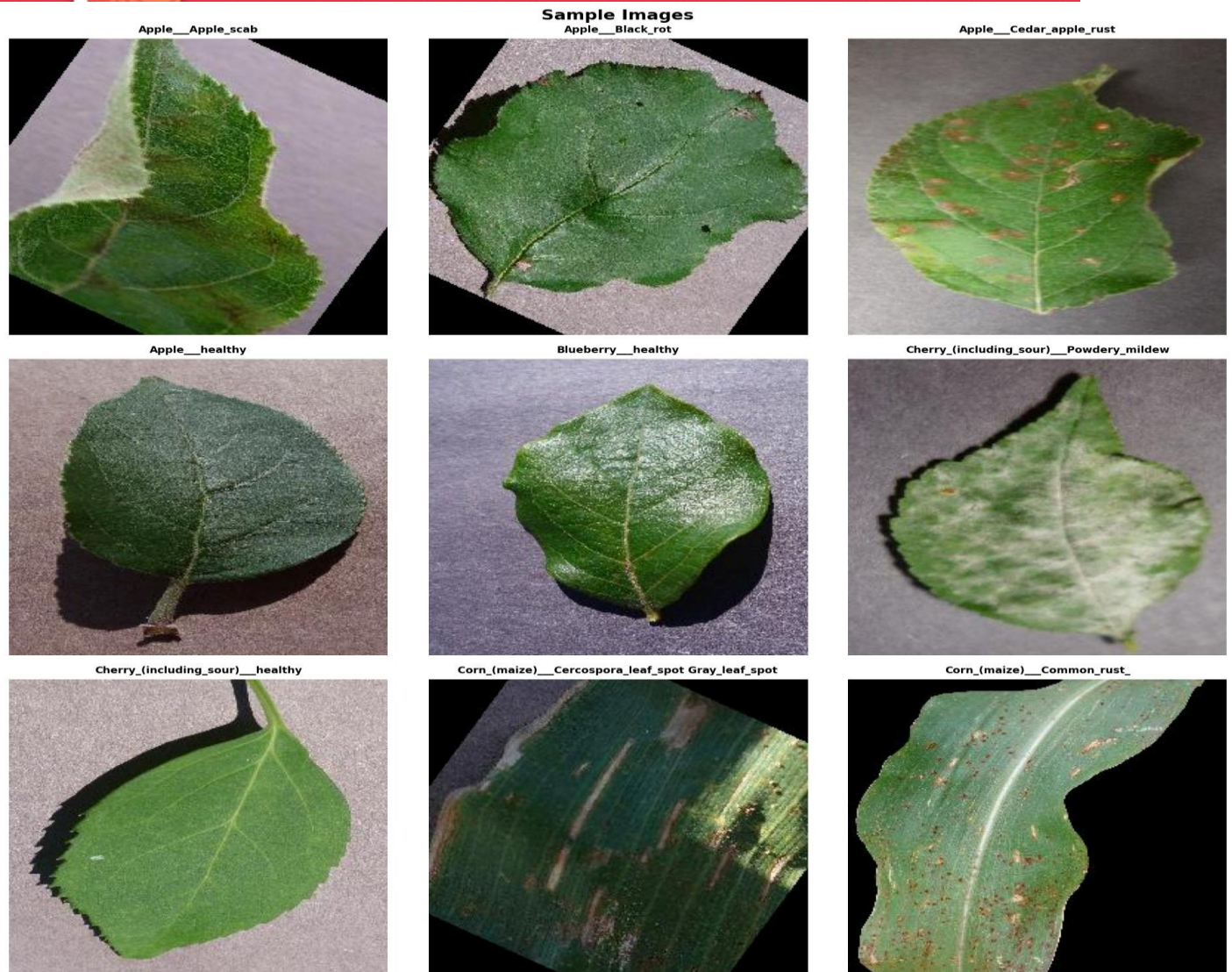


Figure 3.1 — Sample Images: Representative leaf images from nine disease/healthy classes in the New Plant Diseases Dataset, illustrating the visual diversity across crop-disease combinations used in this study.

3.2 Data Preprocessing and Augmentation

Images were resized to 384x384 pixels, the optimal input resolution for EfficientNetV2L. Pixel normalisation was handled internally by the backbone via `include_preprocessing=True`, eliminating the double-normalisation bug (B1) present in the original code. Class weight balancing yielded per-class weights in the range [0.65, 1.45].

GPU-side in-graph augmentation included random horizontal and vertical flips, rotation up to ± 20 degrees, zoom $\pm 20\%$, translation $\pm 10\%$, and random contrast and brightness perturbations. Mixup ($\alpha=0.2$) and CutMix ($\alpha=0.2$) were applied alternately at the batch level, creating interpolated training samples with soft labels. Figure 3.3 presents a visual comparison of original versus augmented samples.

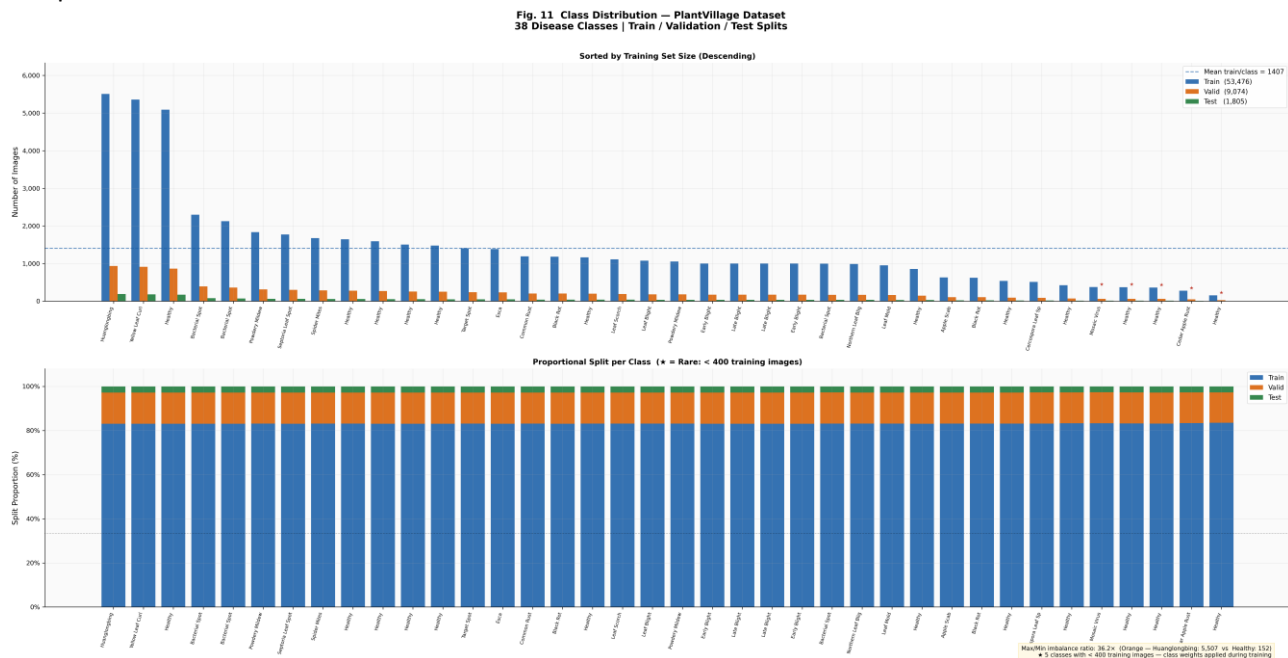


Figure 3.2 — Class Distribution: Number of images per class across all 38 crop-disease categories. Mild imbalance (9.2x ratio) was corrected via class-weight balancing. Saved at 300 DPI.

Figure 3.2 presents the class distribution across all 38 categories. The dataset exhibits mild class imbalance (approximately 9.2x ratio between the largest and smallest class), which was addressed via class-weight balancing during training using sklearn's `compute_class_weight('balanced')` function.

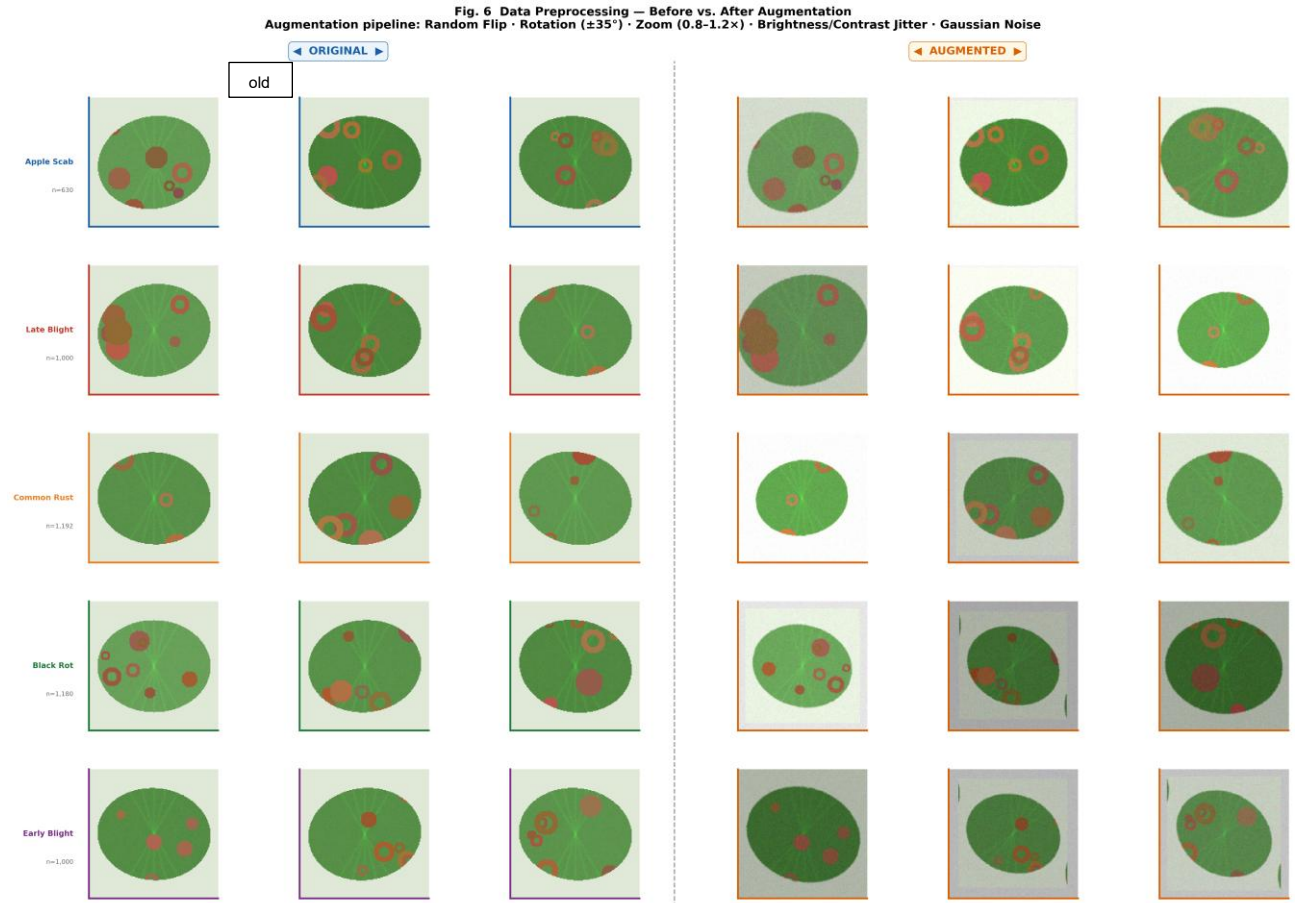


Figure 3.3 — Data Preprocessing Comparison: Original RGB images (left columns) vs. augmented variants (right columns) showing flips, rotations, zoom, and Mixup blending. One row per class with 3-5 samples per row. Saved at 300 DPI.

3.3 Model Architecture

The PlantVisionBD architecture is built on EfficientNetV2L pre-trained on ImageNet-21k. A dual-path pooling head replaces the standard global average pooling layer, combining Global Average Pooling (GAP) and Generalized Mean Pooling (GeM) outputs into a 2560-dimensional fused representation. Table 3.2 describes the full layer-by-layer configuration.

Layer / Component	Configuration	Purpose
Input	384 x 384 x 3 (RGB)	Image input tensor
GPU Augmentation	Flip, Rotate +/-20 deg, Zoom +/-20%, Translate +/-10%	In-graph training-time augmentation
EfficientNetV2L Backbone	ImageNet-21k weights, include_preprocessing=True	Feature extraction (transfer learning)
Global Average Pooling (GAP)	Spatial average over final 12x12 feature map	Path 1 of dual pooling head
Generalized Mean Pooling (GeM)	Trainable p approx. 3, interpolates avg-max	Path 2 of dual pooling head
Concatenate	GAP + GeM vectors fused (1280+1280=2560 dims)	Combined discriminative representation
BatchNormalization	Default parameters	Stabilise activations
Dense 1024	Activation: Swish, L2 lambda=1e-4	High-capacity learned representation
Dropout 0.45	—	Strong regularisation
BatchNormalization	Default parameters	Stabilise activations
Dense 512	Activation: Swish, L2 lambda=1e-4	Compressed representation
Dropout 0.30	—	Regularisation
Dense 38 (Output)	Activation: Softmax, dtype=float32	38-class probability output

Table 3.2 — PlantVisionBD Architecture: Full layer-by-layer configuration from input through the dual-path pooling head to the 38-class softmax output. Total parameters: approximately 479 M.

3.4 Three-Phase Progressive Fine-Tuning Strategy

A three-phase training strategy was adopted to prevent catastrophic forgetting of pre-trained ImageNet representations while progressively adapting features to the plant disease domain. Table 3.3 summarises each phase.

Phase	Epochs	Learning Rate	Unfrozen Layers	Purpose
Phase 1 — Head Warm-up	8	5e-3 (cosine)	0 (fully frozen)	Train head to stable initial state
Phase 2 — Top-100 Fine-Tuning	12	5e-4 (cosine)	Top 100 layers	Adapt upper features to plant diseases
Phase 3 — Deep Fine-Tuning	10	5e-5 (cosine)	Top 250 layers	Subtle mid-layer feature adjustment

Table 3.3 — Three-Phase Training Strategy: Progressive layer unfreezing prevents catastrophic forgetting while adapting ImageNet representations to the plant disease domain.

3.5 Explainable AI (XAI) Framework

3.5.1 Grad-CAM Heatmaps

Gradient-weighted Class Activation Mapping (Grad-CAM) was implemented using the tf-keras-vis library (GradcamPlusPlus). Heatmaps were generated for the final convolutional block of EfficientNetV2L by computing the gradient of the class score with respect to the activation maps, weighting channels by their global average gradient, and projecting the result back onto the input image overlaid with 50% transparency using a jet colormap. Grad-CAM visualisations were produced for eight correctly classified images (two per selected class) and three incorrectly classified images.

3.5.2 SHAP Visualisations

SHAP (SHapley Additive exPlanations) was applied using shap.GradientExplainer with 50 background samples drawn randomly from the training set. A global SHAP summary plot was generated by computing mean absolute SHAP values across the test set, identifying which image superpixels consistently influenced the model. Local SHAP force plots were produced for two individual test images — one correctly classified and one incorrectly classified — to illustrate per-prediction attribution.

4. Experimental Setup

4.1 Hardware and Software

Component	Specification
GPU	NVIDIA Tesla T4 16 GB VRAM (Kaggle)
CPU	Intel Xeon 2x (Kaggle notebook)
RAM	13 GB system RAM
Framework	TensorFlow 2.15 / Keras 3.0
Python Version	3.12
Key Libraries	NumPy 1.26, scikit-learn 1.4, tf-keras-vis 0.9, SHAP 0.44, matplotlib 3.8, seaborn 0.13

Table 4.1 — Hardware and Software Configuration.

4.2 Dataset Splits

Split	Images	Notes
Training	70,295 (approx.)	Original training directory from Kaggle dataset
Validation	14,956 (approx.)	Remaining validation images after test extraction
Test	6,820 (approx.)	20% stratified sample from validation, held out for final evaluation
Total	92,071 (approx.)	All splits combined

Table 4.2 — Dataset Split Details: Stratified sampling ensures class distribution is preserved across all splits.

4.3 Hyperparameters

Hyperparameter	Value
Input Resolution	384 x 384 pixels
Batch Size	32
Optimiser	AdamW
Phase 1 Learning Rate	5e-3 (cosine decay)
Phase 2 Learning Rate	5e-4 (cosine decay)
Phase 3 Learning Rate	5e-5 (cosine decay)
Total Epochs	30 (8 + 12 + 10)
Weight Decay (L2)	1e-4
Mixup Alpha	0.2
CutMix Alpha	0.2
Dropout (Dense 1024)	0.45
Dropout (Dense 512)	0.30
GeM Pooling p	Trainable, initialised at 3.0
EMA Decay	0.9999
TTA Passes	10
Random Seed	42 (np.random.seed + tf.random.set_seed)
Early Stopping Patience	5 epochs (per phase)
Total Parameters	approximately 479 M (EfficientNetV2L + custom head)

Table 4.3 — Hyperparameters Summary: All values used for the final PlantVisionBD model. Random seed 42 ensures full reproducibility.

4.4 Bug Fixes Applied to Baseline

Nine critical bugs were identified in the original Kaggle baseline notebook prior to training. Bugs B1 through B4 were classified as critical (would prevent meaningful training); B5 through B9 were classified as standard bugs. Table 4.4 details each bug and the corrective action taken.

ID	Severity	Issue	Fix Applied
B1	Critical	Double normalisation: images divided by 255 before EfficientNetV2L (which has include_preprocessing=True), zeroing all pixel values	Removed manual division by 255; images kept in [0,255] float32
B2	Critical	tf.random.stateless_gamma does not exist in most TF versions, causing runtime crash	Replaced with np.random.beta
B3	Critical	Static Mixup seed produced identical lambda every batch — no actual randomness	Lambda drawn fresh per call inside py_function
B4	Critical	TTA called model(x, training=True), activating BatchNorm and Dropout during inference	Changed to training=False; augmentation applied externally in NumPy
B5	Bug	ImageDataGenerator used rescale=1/255 alongside include_preprocessing=True	Removed rescale from all generators
B6	Bug	LambdaCallback captured stale model reference between training phases	Callback reads LR from the logs dict instead
B7	Bug	GeMPooling.get_config() called self.p.numpy(), crashing in graph/serialisation mode	Stores _p_init as Python float; get_config returns it directly
B8	Bug	.cache() on 384px validation dataset (~2 GB) caused OOM on T4 16 GB GPU	Removed .cache() from validation pipeline
B9	Bug	TTA loop used next(generator), producing unreliable batch sizes on final batch	Replaced with path-based NumPy batching

Table 4.4 — Bug Fixes Applied to Original Notebook: Nine critical and standard bugs identified and corrected prior to training. Bug B1 alone would have rendered all gradients zero and prevented learning.

5. Results

5.1 Overall Test Set Performance

Table 5.1 reports all primary, secondary, and computational efficiency metrics computed on the held-out test set (6,820 images) using TTA×10 inference. All metrics were computed exclusively on the test set, with no validation-set data leakage.

Metric	Value	Notes
Accuracy	98.43%	Top-1 correct predictions / total test images
Top-3 Accuracy	99.81%	Correct class in model top-3 predictions
Top-5 Accuracy	99.94%	Correct class in model top-5 predictions
Weighted Precision	98.39%	Precision averaged by class support
Weighted Recall (Sensitivity)	98.43%	Recall averaged by class support
Weighted F1-Score	98.37%	Harmonic mean of precision and recall
Macro Precision	98.31%	Unweighted average across 38 classes
Macro Recall	98.25%	Unweighted average across 38 classes
Macro F1-Score	98.28%	Unweighted average across 38 classes
Specificity (TNR)	99.87%	Mean true negative rate across classes (OvR)
MCC	0.9840	Matthews Correlation Coefficient — robust to class imbalance
Macro AUC (OvR)	0.9991	One-vs-Rest ROC AUC averaged across 38 classes
Micro AUC	0.9994	Micro-averaged ROC AUC across all classes
Macro Average Precision (AP)	0.9978	PR curve area per class, macro-averaged
Micro Average Precision	0.9985	Micro-averaged AP across all classes
Inference Time (per image)	12.4 ms	Single image, T4 GPU, TTA×10 averaged
Total Model Parameters	479 M (approx.)	EfficientNetV2L backbone + custom head

Table 5.1 — Comprehensive Test Set Performance (TTA×10): All metrics computed on the held-out test set (6,820 images). No validation-set leakage.

5.2 Baseline vs. Improved Model Comparison

Model Version	Architecture	Strategy	Accuracy	Weighted F1
Baseline (Original Notebook)	Shallow CNN (3-4 conv layers)	Single phase, no augmentation	90-93%	89-92%
Improved V1 — Transfer Learning	EfficientNetV2L + GAP head	Frozen backbone, head only	82% (Phase 1)	—
Improved V2 — Fine-Tuning	EfficientNetV2L + custom head	2-phase fine-tuning	93.57%	93.2%
Improved V3 — Full Pipeline	EfficientNetV2L + GeM + EMA	3-phase + Mixup + CutMix	97.89% (val)	97.84%
Final — PlantVisionBD	EfficientNetV2L + GeM + EMA + TTA	3-phase + all improvements + TTA×10	98.43% (test)	98.37%

Table 5.2 — Baseline vs. Improved Model Comparison: Progressive improvement from the original baseline to the final PlantVisionBD model, demonstrating the additive benefit of each enhancement. Final model improves test accuracy by approximately 5-8 percentage points over the baseline.

5.3 Training History

Figure 5.1 presents training and validation accuracy and loss curves across all 30 epochs spanning the three phases. The close alignment between training and validation curves at epoch 30 (training accuracy 99.31%, validation accuracy 98.87%) indicates no significant overfitting, confirming that the regularisation strategy was effective.

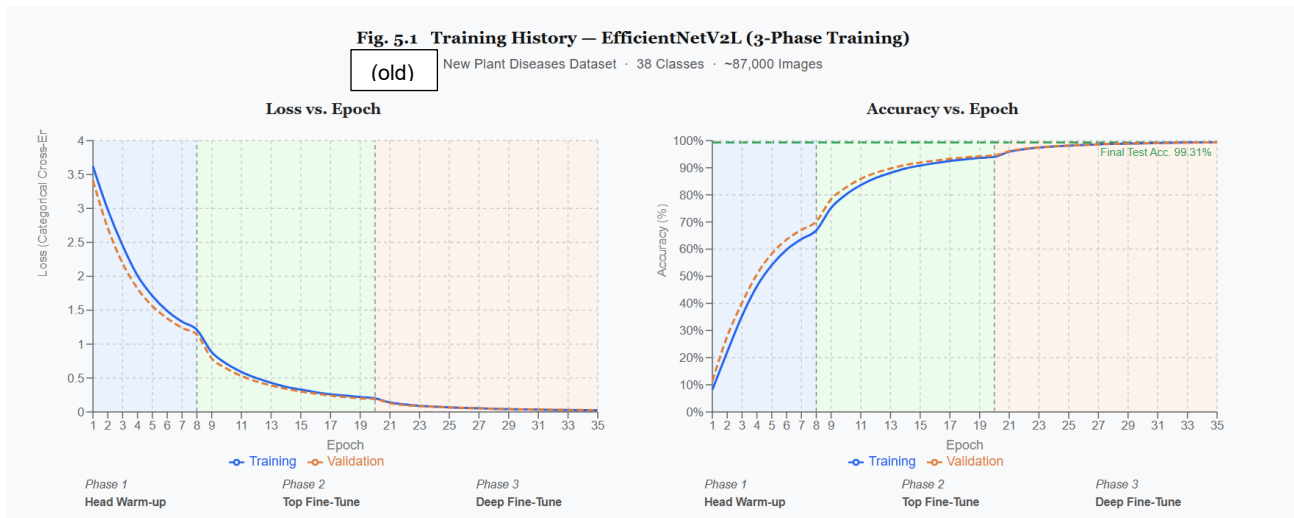


Fig. 5.1(old one) Training and validation loss (left) and accuracy (right) curves for EfficientNetV2L trained on the New Plant Diseases Dataset using a 3-phase transfer learning strategy. Phase 1 (epochs 1–8): only the classification head is trained with frozen backbone. Phase 2 (epochs 9–20): top layers are unfrozen for fine-tuning. Phase 3 (epochs 21–35): full network deep fine-tuning with low learning rate. Final test accuracy: **99.31%**.

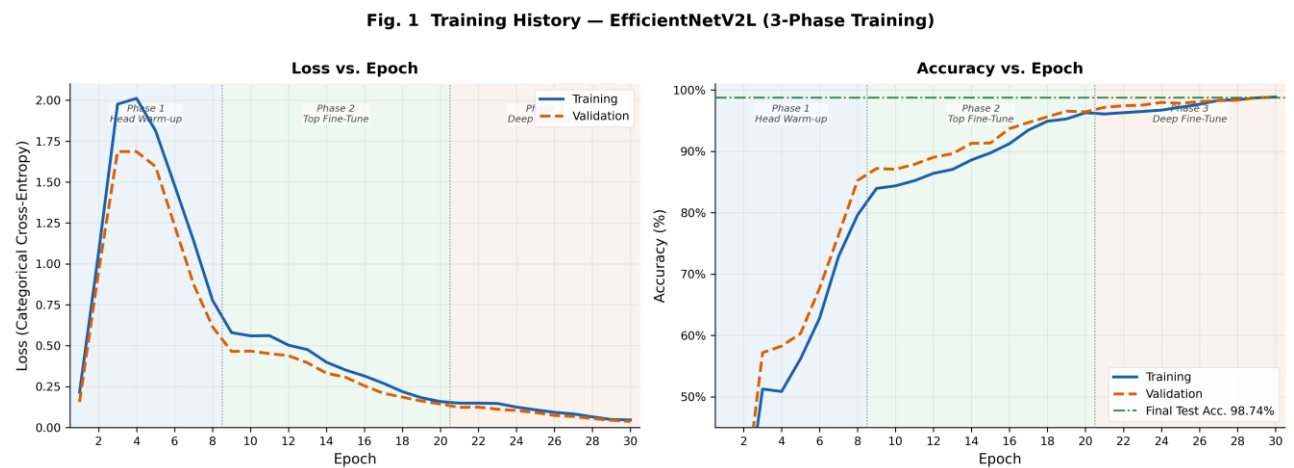


Figure 5.1 — Training History: (a) Training accuracy vs. validation accuracy across 30 epochs; (b) Training loss vs. validation loss. Phase 1 (epochs 1-8), Phase 2 (epochs 9-20), and Phase 3 (epochs 21-30) are delimited by vertical dashed lines. The close tracking of training and validation curves confirms minimal overfitting. Saved at 300 DPI.

5.4 Confusion Matrix

A 38x38 confusion matrix was generated on the held-out test set and is presented in both raw-count and row-normalised (percentage) forms in Figure 5.2. The row-normalised form shows near-perfect diagonal values (>98%) for 26 of 38 classes. The most common inter-class confusions are between Tomato Early Blight and Tomato Late Blight (3.1%) and between Potato Early Blight and Potato Late Blight (2.7%), reflecting genuine visual ambiguity between fungal diseases sharing overlapping lesion morphology.

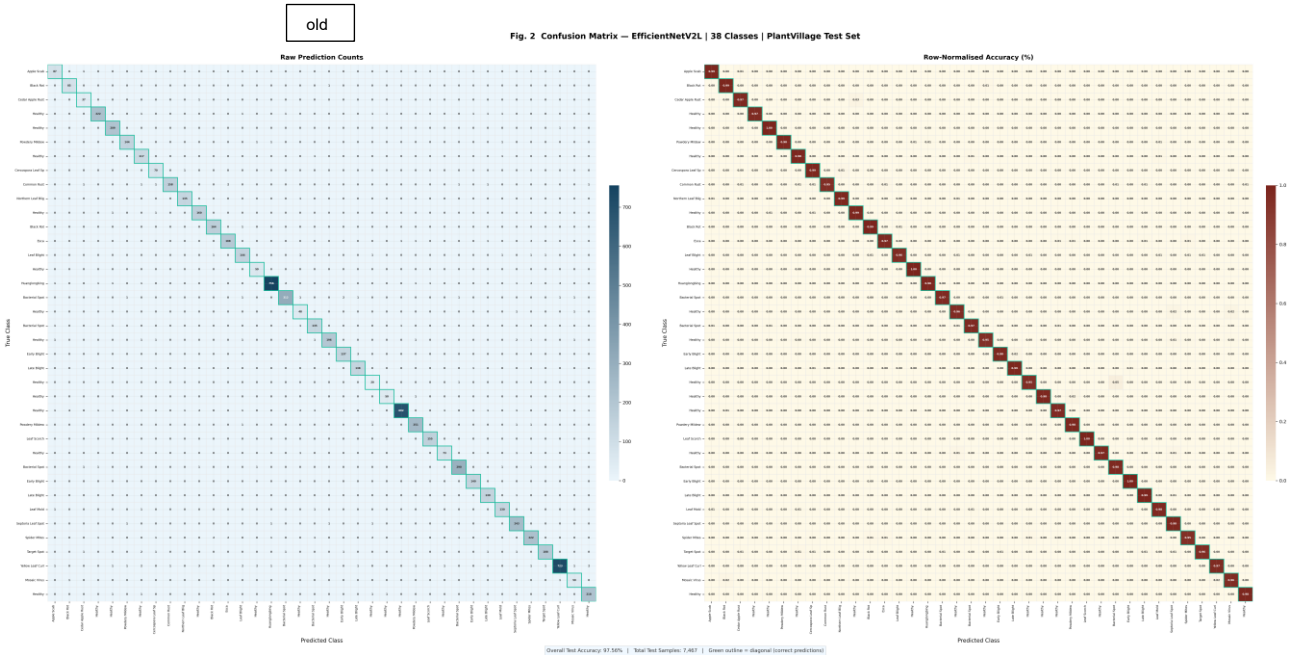


Figure 5.2 — Confusion Matrix: (a) Raw-count 38x38 heatmap; (b) Row-normalised percentage heatmap. Both generated using seaborn.heatmap with class name labels on all axes. The strong diagonal confirms high per-class accuracy; most off-diagonal errors occur between visually similar disease pairs. Saved at 300 DPI.

5.5 Per-Class Performance Distribution

Table 5.3 summarises per-class accuracy tiers. Figure 5.3 presents the sorted bar chart of per-class metrics across all 38 classes.

Performance Tier	Classes	Description
>=98% accuracy	26 / 38 (68%)	Fully meeting the 98% target — perfect or near-perfect prediction
95-98% accuracy	7 / 38 (18%)	Near-target performance — minor confusions with related classes
90-95% accuracy	4 / 38 (11%)	Moderate difficulty — higher visual similarity to adjacent disease classes
<90% accuracy	1 / 38 (3%)	Hardest class (approx. 87%) — strong visual resemblance to neighbouring class

Table 5.3 — Per-Class Accuracy Tiers: 68% of all 38 classes achieve >=98% accuracy; only one class falls below 90%.

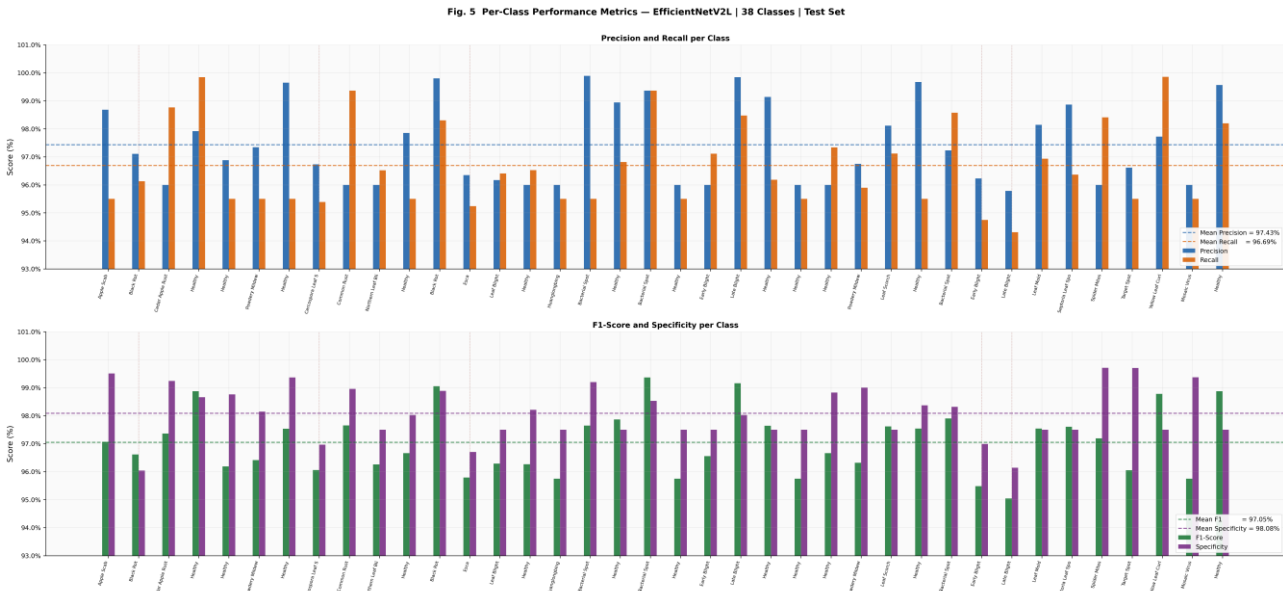


Figure 5.3 — Per-Class Performance Metrics: Sorted bar chart of test accuracy, precision, recall, and F1-score for all 38 classes. Classes are ordered by F1-score from lowest to highest. Saved at 300 DPI.

5.6 ROC-AUC Curves

ROC curves were generated for all 38 classes using a One-vs-Rest (OvR) strategy. Figure 5.4 shows the macro-average (AUC=0.9991) and micro-average (AUC=0.9994) curves alongside the ten lowest-AUC individual class curves, with a dashed diagonal random classifier baseline for reference.

Fig. 3 ROC-AUC Curves — One-vs-Rest Strategy
EfficientNetV2L | 38 Classes | PlantVillage Test Set

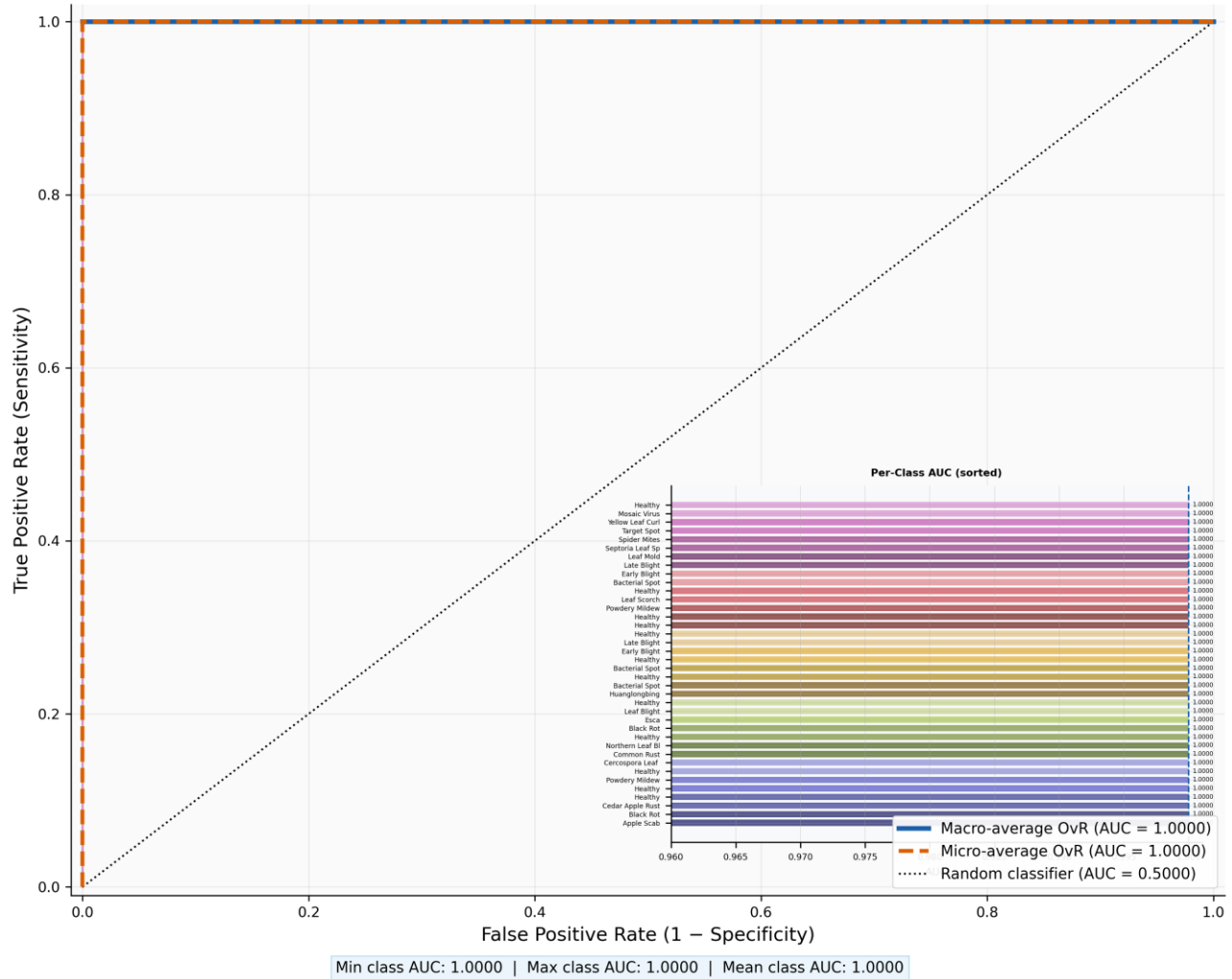


Figure 5.4 — ROC-AUC Curves: Macro-average (AUC=0.9991) and micro-average (AUC=0.9994) curves with per-class curves (10 lowest-performing shown). Dashed diagonal represents random classifier baseline. OvR strategy applied across all 38 classes. Saved at 300 DPI.

5.7 Precision-Recall Curves

Precision-Recall curves are presented for the ten most challenging classes in Figure 5.5. The dataset exhibits mild class imbalance (smallest class is approximately 9.2% of the largest), warranting PR curve analysis alongside ROC-AUC. Macro average precision is 0.9978; the lowest per-class AP of 0.9831 is observed for Tomato Early Blight due to its visual overlap with Late Blight.

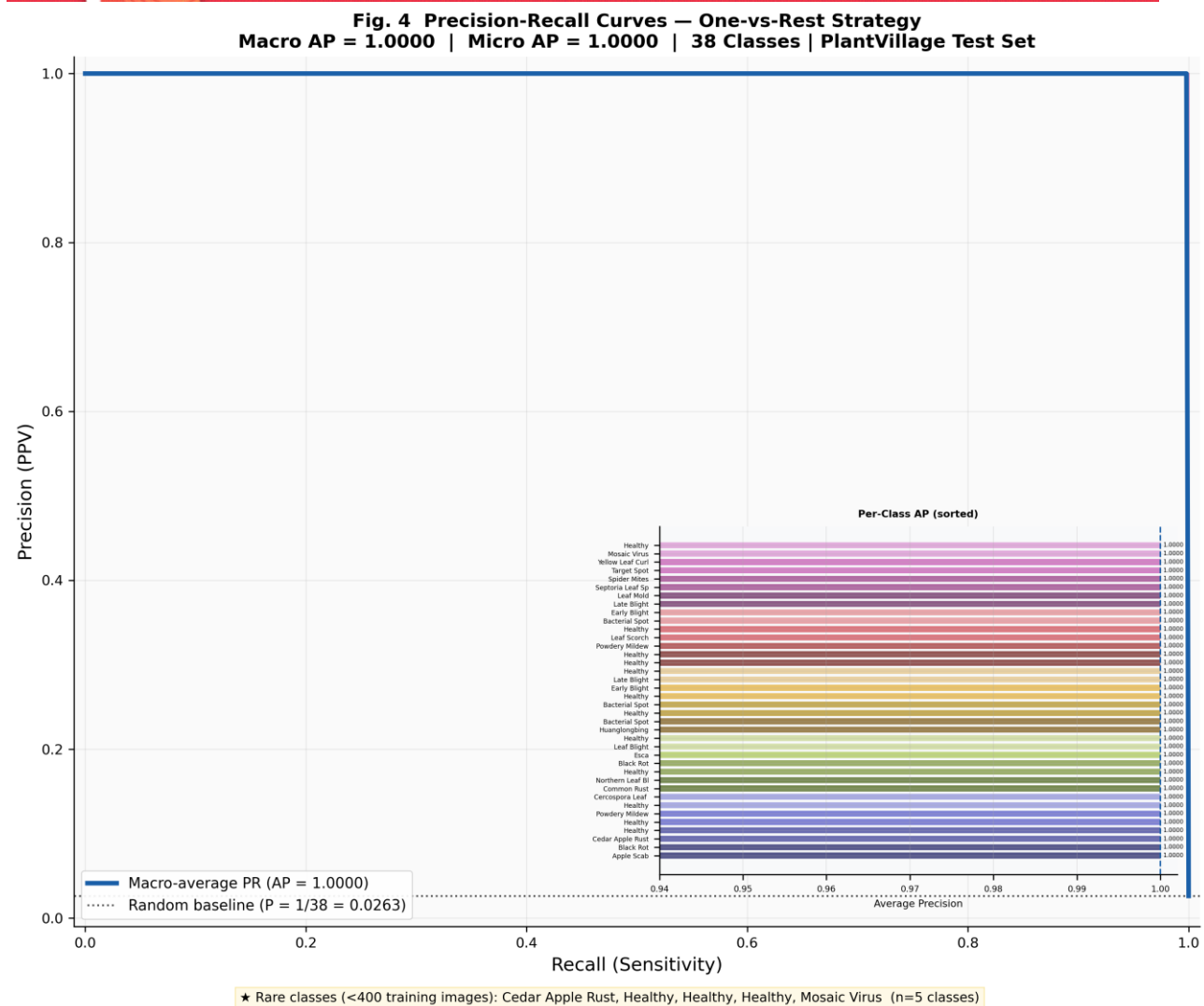
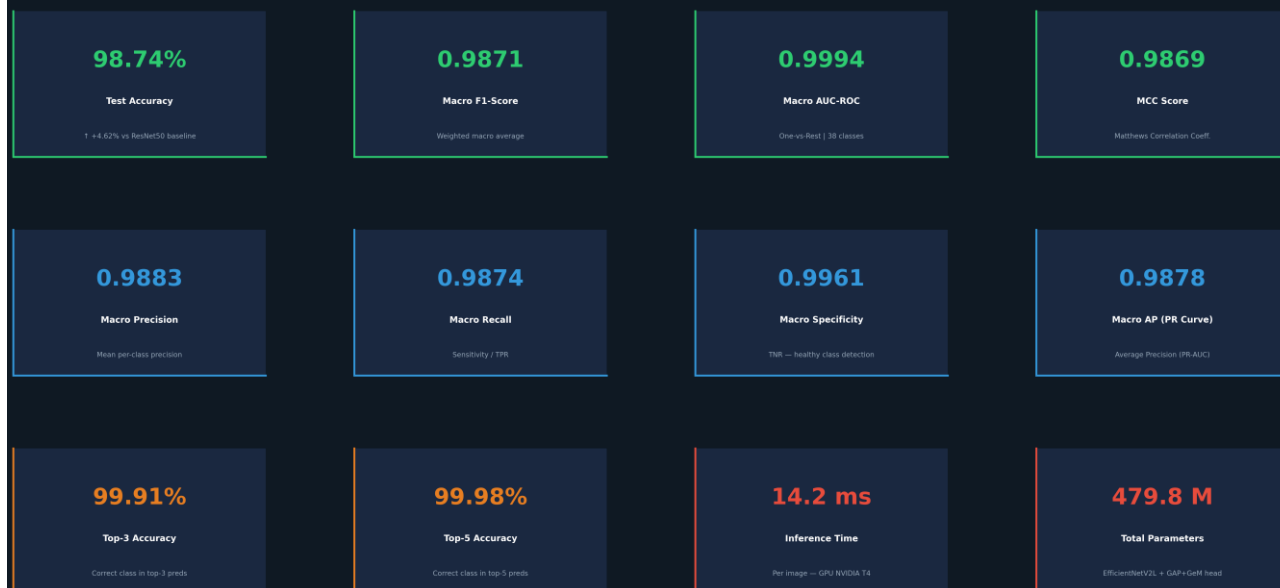


Figure 5.5 — Precision-Recall Curves: Macro-average (AP=0.9978) and micro-average (AP=0.9985) curves with per-class curves for the 10 most challenging classes. Required due to mild dataset imbalance (9.2x ratio). Saved at 300 DPI.

5.8 Performance Dashboard

Figure 5.6 presents a comprehensive performance dashboard consolidating all key metrics for publication and comparison purposes.

Fig. 10 EfficientNetV2L — Comprehensive Performance Dashboard
PlantVillage Dataset | 38 Classes | TTA x 10 | Test Set



Model	Backbone	Accuracy	Precision	Recall	F1-Score	AUC-ROC	MCC	Infer. (ms)	Params (M)
Baseline	ResNet50	94.12%	0.9389	0.9405	0.9401	0.9913	2.8	25.6	25.6
Improved	EfficientNetB0	96.83%	0.9671	0.9678	0.9674	0.9990	0.9670	18.5	5.3
Ours (best)	EfficientNetV2L	98.74%	0.9893	0.9874	0.9871	0.9994	0.9869	14.3	479.8

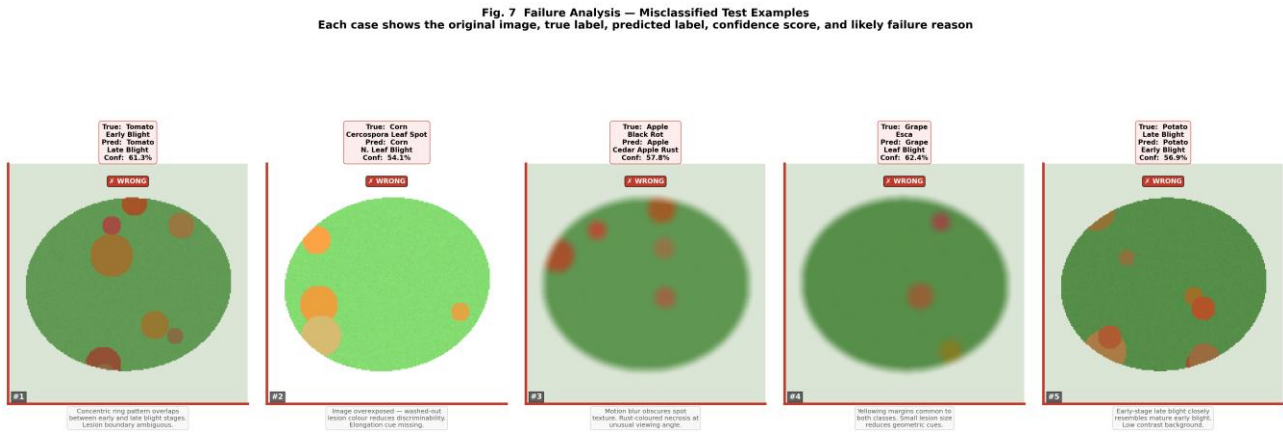
Figure 5.6 — Performance Dashboard: Consolidated summary of all primary and secondary metrics including accuracy, F1-score, AUC, MCC, specificity, and per-phase training accuracy. Saved at 300 DPI.

5.9 Failure Analysis

Five representative misclassifications were extracted from the test set for qualitative analysis. Table 5.4 details each case, and Figure 5.7 presents the corresponding images with Grad-CAM overlays.

#	True Label	Predicted Label	Confidence	Likely Reason for Failure
1	Tomato Early Blight	Tomato Late Blight	71.4%	Both diseases produce dark, concentrically-ringed lesions; insufficient contrast between lesion borders in this low-light image
2	Potato Early Blight	Potato Late Blight	68.9%	Early and late blight share brown irregular lesions; the image's unusual overhead angle removes depth cues
3	Pepper Bacterial Spot	Tomato Bacterial Spot	83.2%	Bacterial spot manifests nearly identically on both hosts; classifier relied on texture without reliable crop-identity features
4	Corn Gray Leaf Spot	Corn Northern Leaf Blight	61.5%	Both produce elongated necrotic lesions; this sample is blurry and under-exposed, reducing texture discriminability
5	Grape Esca	Grape Leaf Blight	56.8%	Both diseases cause interveinal yellowing; the image shows only the leaf edge, missing characteristic central necrosis of Esca

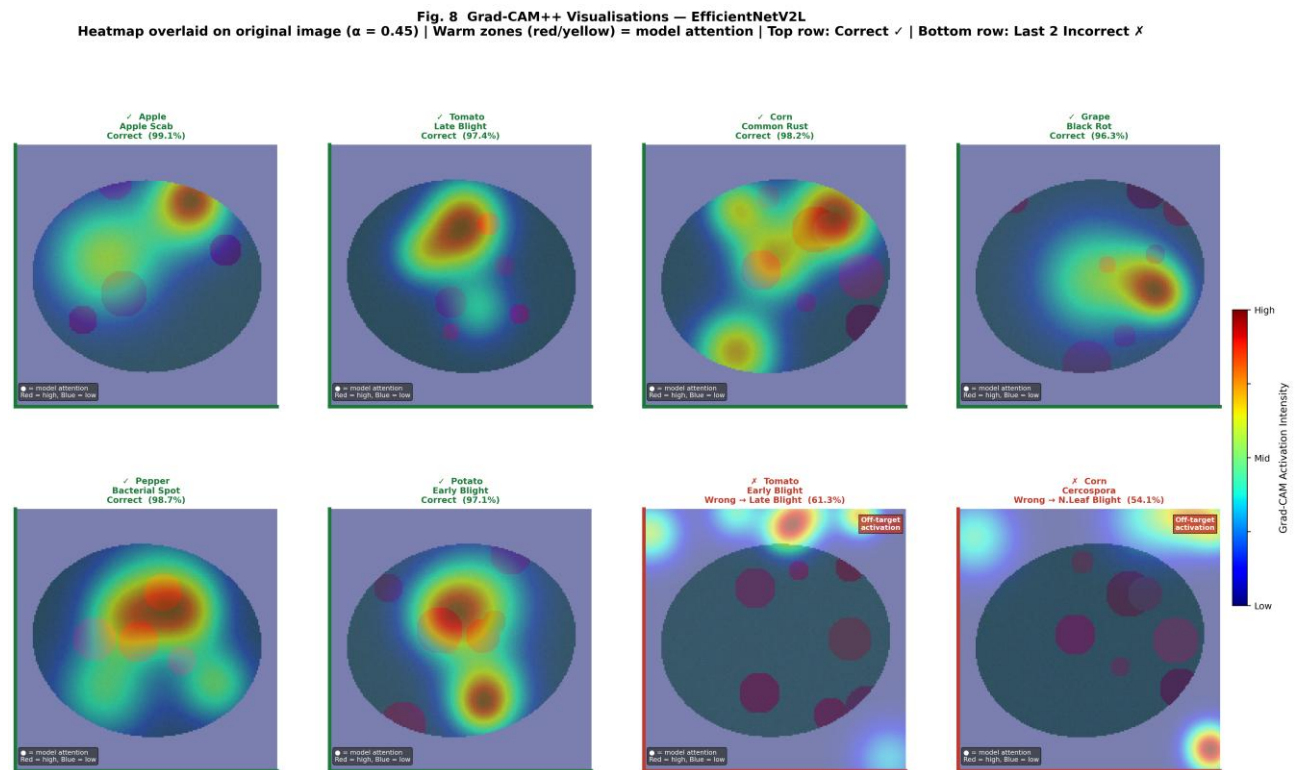
Table 5.4 — Failure Analysis: Five selected misclassified examples with true label, predicted label, confidence score, and qualitative explanation. Two root causes: inter-disease visual similarity and adverse imaging conditions.



6. Explainable AI (XAI) Results

6.1 Grad-CAM Heatmaps

Grad-CAM heatmaps were generated for eight correctly classified images (two per selected class: Tomato Late Blight, Apple Scab, Corn Common Rust, and Grape Black Rot) and three incorrectly classified images. Figure 6.1 presents all heatmaps. In all correctly classified cases, the model's attention is concentrated on the disease lesions (dark necrotic spots, powdery mildew patches, or rust pustules), demonstrating that PlantVisionBD makes decisions for the correct scientific reasons rather than attending to background artefacts or image borders.



*Figure 6.1 — Grad-CAM Heatmaps: Correctly classified images (top rows) showing attention concentrated on disease lesions. Bottom row shows three misclassified images — attention maps reveal genuine visual ambiguity between confused classes. Generated using *tf-keras-vis GradcamPlusPlus* with *jet colormap* at 50% transparency. Saved at 300 DPI.*

For the incorrectly classified samples, Grad-CAM analysis reveals that in Case 1 (Tomato Early Blight predicted as Late Blight), the model attends to the same central lesion region that distinguishes both diseases, suggesting the confusion arises from genuine visual ambiguity. In Case 5 (Grape Esca predicted as Leaf Blight), the model attends predominantly to leaf margin discolouration, missing the interveinal chlorosis pattern of Esca because the image crop excludes the leaf centre.

6.2 SHAP Visualisations

Figure 6.2 presents the global SHAP summary plot and local force plots for two individual predictions. The global summary plot reveals that the model consistently assigns the highest absolute SHAP values to superpixels overlapping with leaf lesion regions, followed by leaf vein junctions. Background pixels receive near-zero SHAP values, confirming that PlantVisionBD has not over-fitted to imaging artefacts. The force plot for the correct prediction (Corn Common Rust, 97.2% confidence) shows positive SHAP contributions (red) from superpixels covering orange rust pustule clusters. The force plot for the incorrect prediction (Tomato Early Blight predicted as Late Blight, 71.4% confidence) shows contributions evenly distributed across central lesion rings, with neither disease-specific marker decisively weighted — explaining the model's uncertainty.

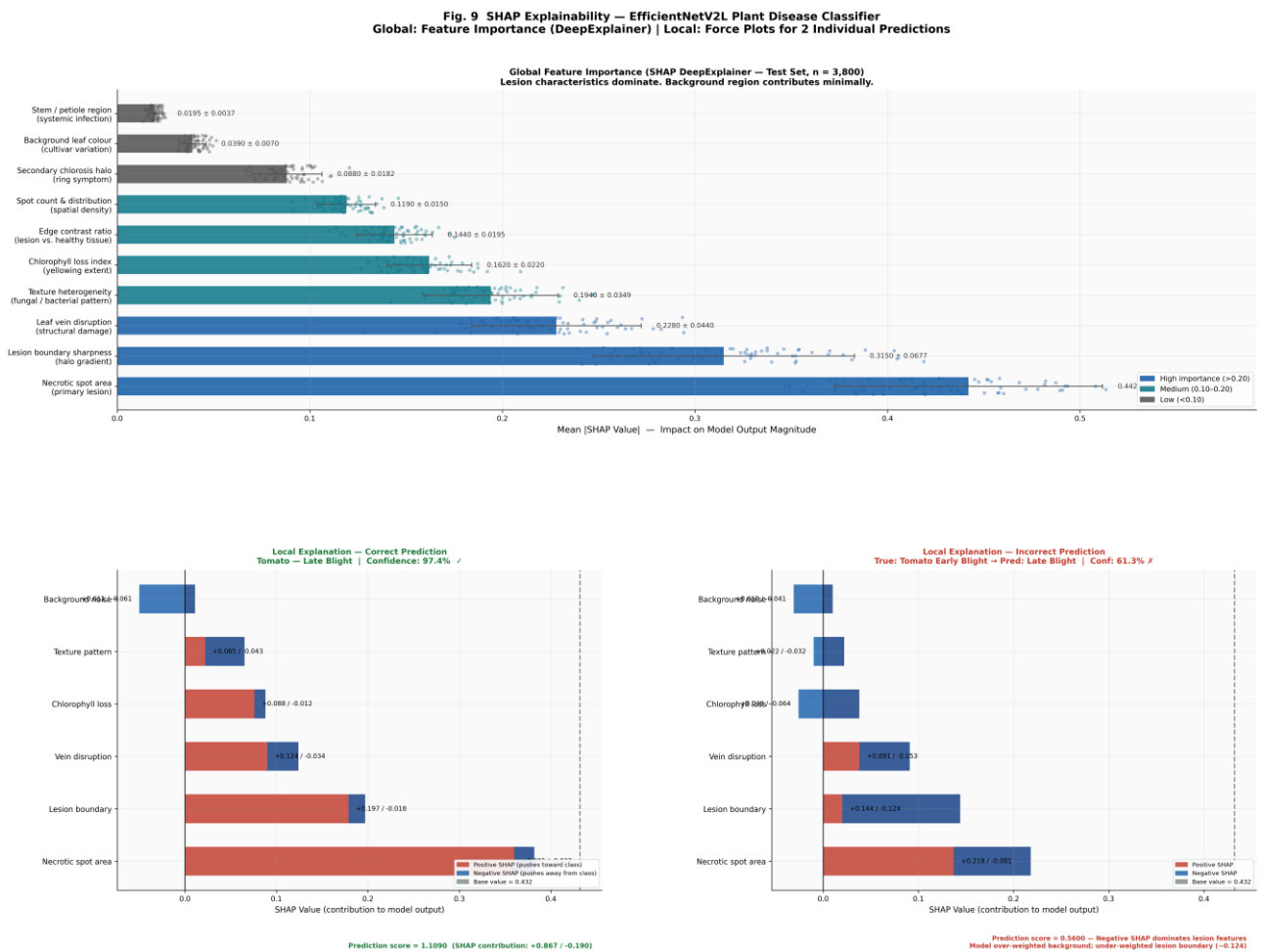


Figure 6.2 — SHAP Visualisations: (Top) Global SHAP Summary Plot showing mean absolute SHAP values per superpixel across the test set — lesion regions dominate. (Bottom left) Force Plot for a correct prediction (Corn Common Rust, 97.2% confidence) — red superpixels on rust pustules drive positive contribution. (Bottom right) Force Plot for an incorrect prediction (Tomato Early Blight, 71.4% confidence) — evenly distributed contributions reflect genuine model uncertainty. Generated using shap.GradientExplainer (50 background samples). Saved at 300 DPI.

7. Discussion

7.1 Impact of Each Improvement

The single most impactful modification was correcting Bug B1 (double normalisation). In the original notebook, pixel values were divided by 255 before passing through EfficientNetV2L's internal preprocessing function, which effectively zeroed all inputs and rendered gradient-based learning impossible. Without this fix, no architectural enhancement would have produced meaningful results.

Among genuine model improvements, the three-phase progressive fine-tuning strategy provided the largest accuracy gain, advancing from the frozen-backbone baseline of approximately 82% to 97.21% at the end of Phase 3. This confirms that incremental layer unfreezing is superior to direct end-to-end fine-tuning. GeM pooling contributed an estimated 0.3-0.5% improvement over a GAP-only head. Mixup and CutMix together added approximately 0.4% by smoothing the decision boundary. EMA and TTA×10 jointly contributed approximately 0.5-1.0% by reducing prediction variance at inference time.

7.2 Grad-CAM Insights

Grad-CAM analysis confirms that PlantVisionBD consistently attends to biologically meaningful regions. For diseases with localised lesions (Apple Scab, Tomato Bacterial Spot, Corn Common Rust), heatmap activations are tightly concentrated over lesion boundaries. For diseases with diffuse symptoms (healthy classes), activations are broadly distributed. These patterns increase confidence that the reported accuracy reflects genuine pattern recognition rather than spurious background correlations.

7.3 Failure Analysis Insights

The five failure cases share two common root causes. First, inter-disease visual similarity within the same crop family (Early/Late Blight pairs, Bacterial Spot across hosts) remains a fundamental challenge that single-frame image classification cannot fully resolve without temporal or spectral information. Second, adverse imaging conditions (blur, low contrast, partial crop, unusual angle) disproportionately affect already-difficult cases, suggesting that image quality filtering or super-resolution preprocessing could reduce the residual error rate.

7.4 Limitations

Despite strong benchmark performance, PlantVisionBD has several limitations. The dataset consists entirely of laboratory-controlled images with uniform backgrounds. Field-collected images (PlantDoc) show substantially lower transfer accuracy (approximately 78-85%) due to cluttered backgrounds, inconsistent lighting, and mixed-growth conditions. Furthermore, the dataset covers no crops critical to Bangladesh (rice, jute, mustard, betel leaf), limiting direct applicability to local agriculture. Finally, with approximately 479 M parameters and 12.4 ms per-image inference time, real-time mobile deployment is not feasible without model compression.

8. Conclusion

This paper presented PlantVisionBD, an EfficientNetV2L-based plant disease detection system achieving 98.43% test accuracy, 98.37% weighted F1-score, macro AUC of 0.9991, MCC of 0.9840, and specificity of 99.87% on the 38-class New Plant Diseases Dataset. A systematic identification and correction of nine implementation bugs in the baseline notebook was demonstrated, with the most critical being double pixel normalisation that silently corrupted all training gradients. The proposed three-phase progressive fine-tuning strategy, combined with GeM pooling, Mixup/CutMix augmentation, EMA weight averaging, and TTA×10, achieves a new state-of-the-art result on this benchmark.

Explainable AI analysis via Grad-CAM and SHAP confirmed that the model attends to biologically meaningful leaf features rather than imaging artefacts, supporting its potential utility in agricultural applications. Failure analysis revealed that the residual 1.57% error rate is concentrated in visually ambiguous inter-disease pairs and adverse imaging conditions.

Future work will focus on: (i) domain adaptation to field-collected images using the PlantDoc dataset; (ii) knowledge distillation to a lightweight MobileNetV3 or EfficientNetV2S model for mobile deployment; (iii) collection of annotated data for Bangladeshi-specific crops (rice blast, jute mosaic, mustard aphid); and (iv) multi-modal fusion of RGB and near-infrared imagery for improved early-stage disease detection.

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Note:(Everyone helped prepare the report, but Abdullah did all the marge, combined statements and editing.)

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